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Launching VPC Resources

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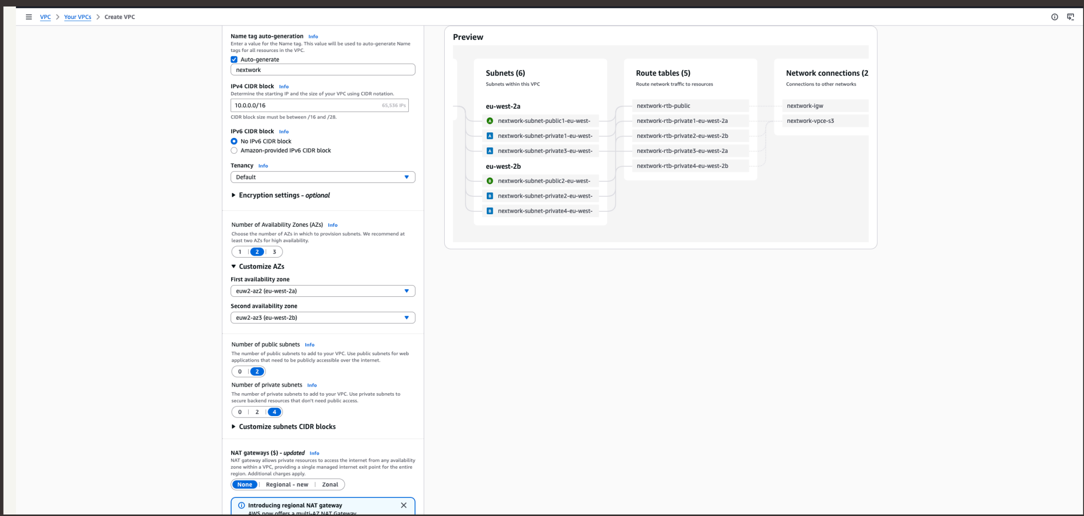

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a service that lets you create your own isolated private network within AWS where you can launch and manage cloud resources. It is useful because it gives you full control over your network environment, allowing you to decide which resources are publicly accessible and which are kept private, configure your own IP address ranges, and control traffic using security groups and network ACLs, making it possible to build a secure and structured network that mirrors how a real world company network would operate.

How I used Amazon VPC in this project

I used Amazon VPC to build a complete network infrastructure by creating a VPC, setting up both public and private subnets, and launching EC2 instances into each one. I configured route tables, security groups and network ACLs to control how traffic flows between the instances and the internet, and finished by using the VPC wizard to see how the entire setup could be recreated in a fraction of the time compared to building everything manually.

One thing I didn't expect in this project was...

One thing I did not expect in this project was how much the security group configuration mattered when setting up communication between the public and private EC2 instances. I did not realise that instead of specifying an IP address as the source, you could reference another security group directly, which was a much more elegant and flexible way of controlling access between instances within the same VPC.

This project took me...

This project took me roughly an hour to complete

Setting Up Direct VM Access

Directly accessing a virtual machine means connecting to it through a terminal and controlling it by typing commands, rather than interacting with it through a visual interface. It is similar to being sat in front of the computer itself, except you are doing it remotely over the internet using a secure connection, allowing you to run programs, manage files and configure settings directly on the machine.

SSH is a key method for directly accessing a VM

SSH stands for Secure Shell. SSH traffic means the data being sent between your computer and a remote server through a secure, encrypted connection. It is the protocol that allows you to remotely log in to and control another computer, such as an EC2 instance, safely over the internet by ensuring that no one else can intercept or read the data being transmitted.

To enable direct access, I set up key pairs

Key pairs are a secure login method used to access an EC2 instance. They work by generating two keys, a public key that AWS holds and a private key that you download and keep on your own computer. When you try to connect to your instance, AWS checks that your private key matches the public key, and only then grants you access. This is much more secure than using a simple username and password.

A private key's file format refers to the type of file that your private key is saved as, which determines how it can be used to connect to your EC2 instance. Different tools use different formats, for example .pem files are used by terminal based tools like SSH on Mac and Linux, while .ppk files are used by Windows tools like PuTTY. My private key's file format was .pem.

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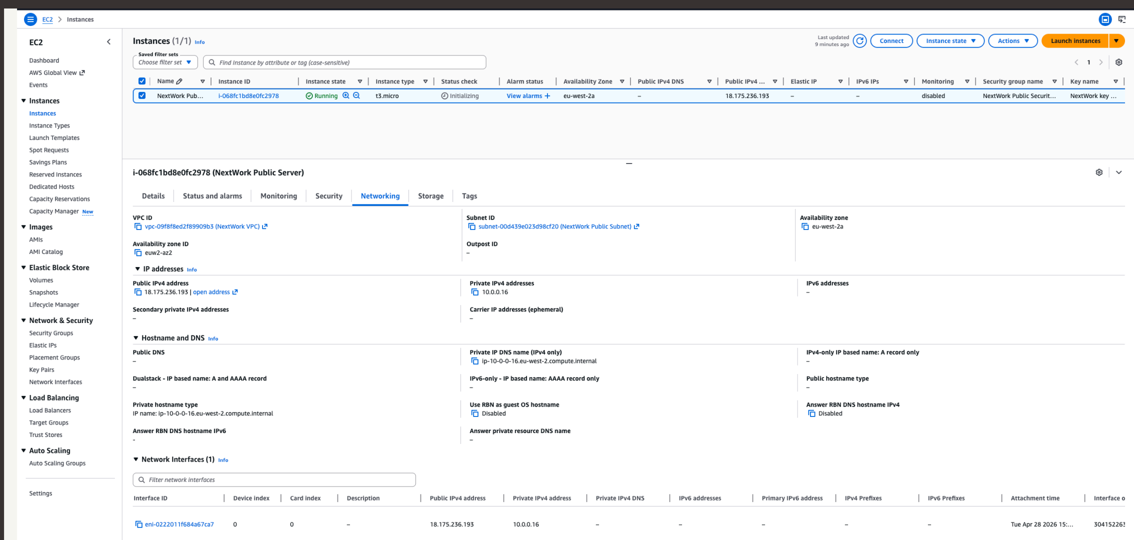
Launching a public server

I had to change my EC2 instance's networking settings by navigating to the Network Settings section during the EC2 instance setup, where I selected my custom VPC instead of the default one, chose my public subnet to ensure the instance would be publicly accessible, and enabled auto-assign public IP so that the instance would be given a public IP address that could be reached over the internet.



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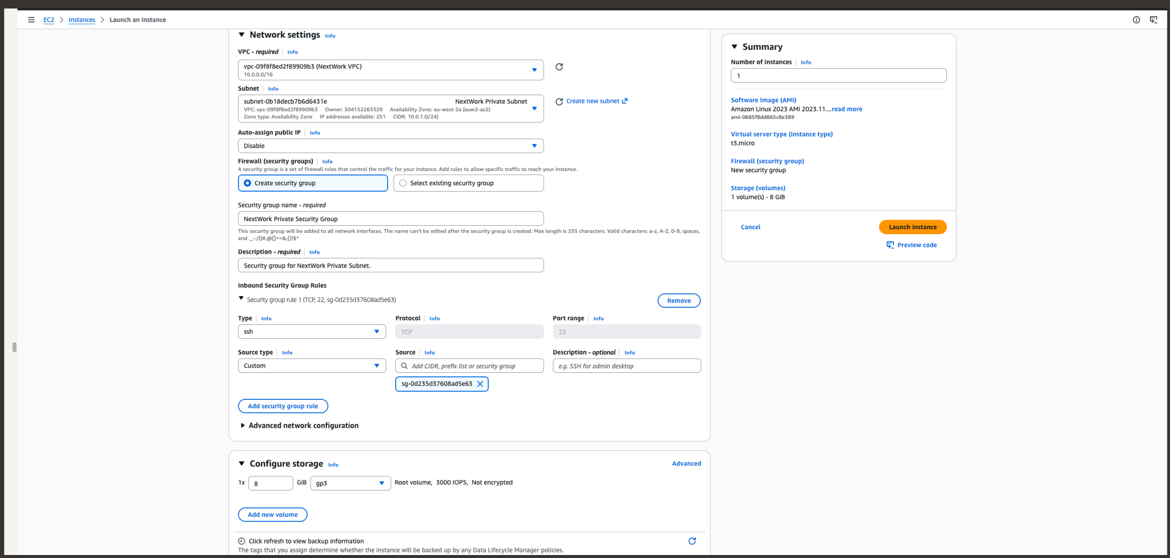
Launching a private server

My private server has its own dedicated security group because it has different access requirements to the public server. The public server needs to accept traffic from the internet, whereas the private server should only accept traffic from within the VPC, specifically from the public server. Having a separate security group allows me to set rules that restrict access to the private instance so that it can only be reached by the public EC2 instance and not directly from the internet, keeping it secure.

My private server's security group's source is the public server's security group, which means that only traffic coming from the public EC2 instance is allowed to reach the private EC2 instance. Rather than specifying a specific IP address, referencing the security group itself ensures that any instance associated with the public security group can communicate with the private server, making it a more flexible and secure way of controlling access between the two instances.

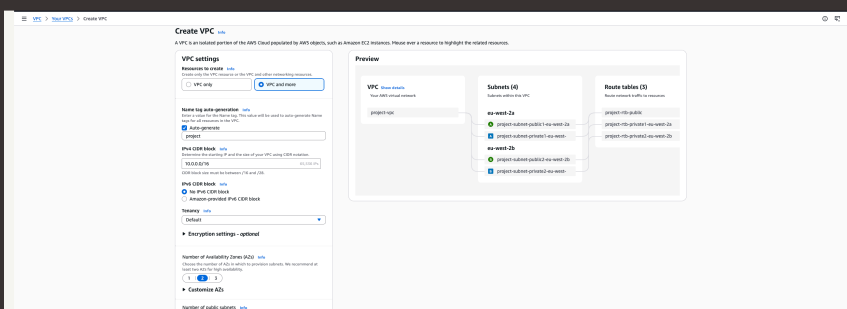


Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, I used the AWS VPC wizard which allowed me to create all of the necessary resources including the VPC, subnets, route tables and internet gateway all at once in a single streamlined process, rather than creating each resource individually as I had done in the previous projects. This made the whole setup significantly faster and demonstrated how the wizard can be a much more efficient option once you understand what each component does.

A VPC resource map is a visual diagram within the AWS console that shows all of the resources inside your VPC and how they are connected to each other. It gives you a clear overview of your subnets, route tables, internet gateways and other components laid out in a way that makes it easy to understand the structure and traffic flow of your entire network at a glance, without having to navigate through each resource individually.

My new VPC has a CIDR block of 10.0.0.0/24. It is possible for my new VPC to have the same IPv4 CIDR block as my existing VPC because each VPC is a completely isolated private network within AWS. They do not communicate with each other by default, so having the same IP address range does not cause any conflicts, in the same way that two separate office buildings could both use the same internal phone extension numbers without any issues because they operate independently of each other.

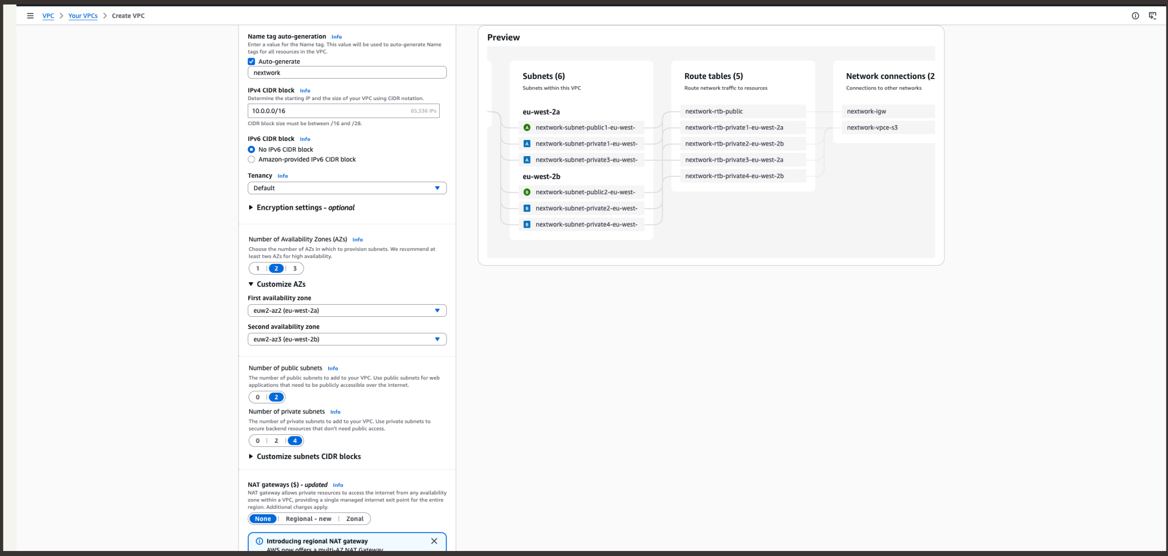


Speeding up VPC creation

Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options: 0 or 1. This was because the CIDR block I used for my VPC is 10.0.0.0/24, which only provides 256 IP addresses in total. With the IP addresses already being divided between the public and private subnets, there was only enough space to support one public subnet within the available address range.

The set up page also offered to create NAT gateways, which are devices that allow resources inside a private subnet to access the internet for things like downloading updates or making outbound requests, without exposing them to inbound traffic from the internet. This means a private EC2 instance can reach out to the internet when it needs to, while still remaining protected and unreachable from the outside world.



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